CV

AI CAD-to-Cost Analysis

CostVision · Powered by Claude AI · Manufacturing Intelligence Platform

AI-POWERED CAD ANALYSIS REPORT

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Geometry to Process to Cost · Fully AI-reasoned

53

Automotive Bumper Fascia (BUMPER.stp)

1691 × 647 × 528 mm · 2059.9 cm³ · Al 5.562 kg / Steel 16.170 kg

Manufacturability: 53/100 · Confidence: Low

§1 — Geometry & Part Summary

Bounding Box	**1690.9 × 647.4 × 528.1 mm**	Surface Area	**16261.7 cm²**
Volume	**2059.90 cm³**	Weight (Al)	**5.562 kg**
Weight (Steel)	**16.170 kg**	Weight (Plastic)	**2.163 kg**

§2 — Detected Features

Feature Type	Count	Significance	Description
Free-form B-spline surfaces	492	High	492 of 498 faces are B-spline — organic styled exterior panel, styled contours and ribs
Undercut faces	4	High	4 undercut faces requiring side-actions or re-orientation of pull direction
Zero-draft faces	2	Medium	2 faces with no draft — risk of drag marks / ejection difficulty
Threaded feature	1	Medium	Thread detected — likely a boss/insert for fastener attachment
Non-uniform thin wall	1	Medium	Wall varies 2.53–3.54mm, consistent with a large exterior IM panel

§3 — Material Analysis

AI-suggested

Primary Material: **PP Impact Copolymer (PP-B) (80% confidence)**

Part geometry (large single-shell exterior styled panel, thin 2.5–3.5mm wall, huge projected area) is classic automotive bumper fascia. PP-B (impact copolymer, often talc/EPDM modified) is the overwhelming industry-standard resin for bumper fascias due to low cost, high impact toughness at low temp, and paintability. Resin choice was UNDECIDED in the geometry data and is selected here from part-type/name inference, not measured geometry.

Alternatives: **PC/ABS Blend (automotive grade) (10%) · ASA (6%) · ABS (4%)**

§4 — Process Recommendations

Process	Commodity	Confidence	Cycle Time (hr)	Reasoning
Injection Moulding	injection_moulding	95%	0.0125	User-forced primary process. Large thin-shell styled exterior part with 492/498 free-form faces and thin near-uniform wall (2.5–3.5mm) is textbook injection-moulded bumper fascia geometry. Low fill ratio (0.0036) reflects a hollow/shell-like outer skin, not a solid casting.
Blow Moulding	blow_moulding	5%	0.0200	Considered due to low fill ratio/hollow shell signature, but rejected — bumper fascias require styled A-surface, ribbed backside, and mounting bosses/threads incompatible with blow moulding; user-forced IM overrides this alternative.

§5 — Manufacturability Risks (Score: 53/100)

Severity	Feature / Area	Description	Recommended Action
High	**4 undercut faces**	Undercuts on the styled surface/backside ribs will require side-action sliders or lifters, adding tool complexity and cost.	Review undercut geometry for re-orientation of pull direction; where unavoidable, add hydraulic/mechanical side-actions.
High	**Large flat/curved panel (1691×647mm)**	Very large surface-to-thickness ratio (thin 2.5–3.5mm wall over ~1.7m span) is highly prone to differential cooling, warpage and panel bow.	Use balanced multi-gate hot-runner fill, uniform mould cooling circuits, and ribbed backside stiffening; consider conformal cooling in high-warp zones.
Medium	**Non-uniform wall thickness (2 . 5 3 – 3 . 5 4 m m , Ã :**	Local thickness variation risks sink marks and uneven shrinkage, especially near boss/thread feature.	Normalise wall thickness where possible; add gussets instead of thick bosses; use core-outs to reduce local mass concentration.
Medium	**Weld lines from large multi-gate fill**	A part this size will require multiple gates (hot runner), creating weld/knit lines that are cosmetic and structural risk zones on a Class-A exterior surface.	Use mould-flow simulation to position gates away from high-visibility A-surface and high-stress mounting zones.
Medium	**Zero-draft faces (2)**	Two zero-draft faces increase ejection friction risk and potential surface drag/scuffing on cosmetic surface.	Add minimum 1–2° draft on all non-critical faces; texture-matched draft angles on Class-A surface.

§7 — AI Indicative Cost Range (model opinion — not engine-calculated)

Net Weight	**1.854 kg**	Material	**mat-pp-impact**
Recommended Process	**machining**	Cycle Time	—
Setup Time	—	Operations	**0 ops**
Operations Detail			

Process-Specific Parameters

Cavities	**1**
Wall Thickness	**2.5 mm**
Mould Cost	**£312,157**
Mould Life	**1,000,000 shots**
Projected Area	**10946.8 cm²**

§8 — AI Analysis Explanation